

Planning Logic

Project Title

Virtual Agent–Driven SLA Breach Awareness & Justification System

Introduction

Planning Logic defines the sequence of activities, decision-making process, and workflow followed during the development of the **Virtual Agent–Driven SLA Breach Awareness & Justification System**. It provides a structured roadmap for transforming project requirements into a fully functional application by organizing each development phase systematically.

The planning process ensures that every stage of the project—from requirement analysis to deployment—is completed efficiently while maintaining quality, reliability, and alignment with project objectives.

Project Development Approach

The project follows a structured software development approach consisting of the following phases:

1. Requirement Analysis
2. System Design
3. Development
4. Testing
5. Deployment
6. Evaluation and Maintenance

Each phase contributes to the successful implementation of the system.

Planning Workflow

Phase 1 – Requirement Analysis

Activities:

- Identify problems in existing SLA management.
- Gather user requirements.
- Define project objectives.
- Analyze business requirements.
- Study existing ITSM workflows.

Output:

- Functional requirements

- Non-functional requirements
 - User requirements
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Phase 2 – System Design

Activities:

- Design system architecture.
- Design database structure.
- Plan user interface.
- Define workflow automation.
- Design Virtual Agent interaction.
- Plan AI integration.

Output:

- System architecture
 - Database design
 - Workflow diagrams
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Phase 3 – Development

Activities:

- Develop frontend interfaces.
- Implement backend services.
- Configure database.
- Integrate ServiceNow workflows.
- Integrate Google Gemini AI.
- Develop dashboards and reports.

Output:

- Functional application modules
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Phase 4 – Testing

Activities:

- Functional testing
- Performance testing

- User Acceptance Testing (UAT)
- Workflow validation
- AI response verification
- Notification testing

Output:

- Verified and stable application
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Phase 5 – Deployment

Activities:

- Deploy application.
- Configure production environment.
- Verify workflows.
- Validate dashboards.
- Test Virtual Agent.

Output:

- Operational system
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Phase 6 – Evaluation & Maintenance

Activities:

- Monitor application performance.
- Review SLA compliance.
- Collect user feedback.
- Resolve issues.
- Plan future enhancements.

Output:

- Improved application performance
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Development Logic

The development process follows this logical sequence:

Requirements Collection



Requirement Analysis



System Design



Database Design



Frontend Development



Backend Development



ServiceNow Workflow Configuration



Google Gemini AI Integration



Dashboard Development



Testing



Deployment



Maintenance

Workflow Logic

The operational workflow of the application follows these steps:

User Login



Incident Creation or Update



Automatic SLA Attachment



Real-Time SLA Monitoring

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SLA Status Classification

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Flow Designer Workflow Trigger

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Notification Generation

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Virtual Agent Interaction

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AI-Based SLA Justification

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User Acknowledgement

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Incident Update

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Dashboard & Report Generation

Decision Logic

The system continuously evaluates the SLA status using predefined rules.

Condition 1

If sufficient SLA time remains:

Status = SAFE

Action:

- Continue monitoring.
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Condition 2

If the remaining SLA time reaches the warning threshold:

Status = AT RISK

Action:

- Notify assigned support agent.
- Launch Virtual Agent guidance.

- Generate AI recommendation.
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Condition 3

If the SLA deadline is exceeded:

Status = BREACHED

Action:

- Require SLA breach justification.
 - Record acknowledgement.
 - Update audit logs.
 - Notify management.
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Resource Planning

The project requires the following resources:

Human Resources

- Project Developer
 - Support Agent
 - Team Lead / Manager
 - Administrator
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Software Resources

- Python
- Flask
- HTML5
- CSS3
- JavaScript
- Chart.js
- SQLite
- ServiceNow
- Flow Designer
- Google Gemini AI
- Visual Studio Code

- GitHub
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Hardware Resources

- Desktop/Laptop
 - Internet Connection
 - Modern Web Browser
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Risk Planning

Potential project risks include:

- AI service unavailability.
- Network connectivity issues.
- Incorrect SLA configuration.
- Workflow execution failures.
- Database inconsistencies.
- User authentication failures.

Mitigation Strategies

- Validate workflows before deployment.
 - Implement secure authentication.
 - Perform regular database backups.
 - Monitor AI service availability.
 - Conduct periodic testing.
 - Maintain audit logs.
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Expected Deliverables

The project is expected to deliver:

- Intelligent SLA monitoring system.
- AI-powered Virtual Agent.
- Automated workflow management.
- SLA governance dashboard.
- Reporting and analytics.
- Secure role-based application.

- Documentation and project reports.
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Benefits of Proper Planning

Effective planning provides several advantages:

- Clear project direction.
 - Better resource utilization.
 - Reduced development risks.
 - Faster implementation.
 - Improved software quality.
 - Efficient testing process.
 - Easier maintenance.
 - Successful project delivery.
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Conclusion

The Planning Logic document provides a structured roadmap for the successful development of the **Virtual Agent–Driven SLA Breach Awareness & Justification System**. By organizing project activities into clearly defined phases, decision points, and workflows, the planning process ensures timely development, efficient resource utilization, proactive risk management, and high-quality software delivery. The logical sequence of analysis, design, implementation, testing, deployment, and maintenance establishes a strong foundation for building a reliable, scalable, and intelligent SLA management solution.